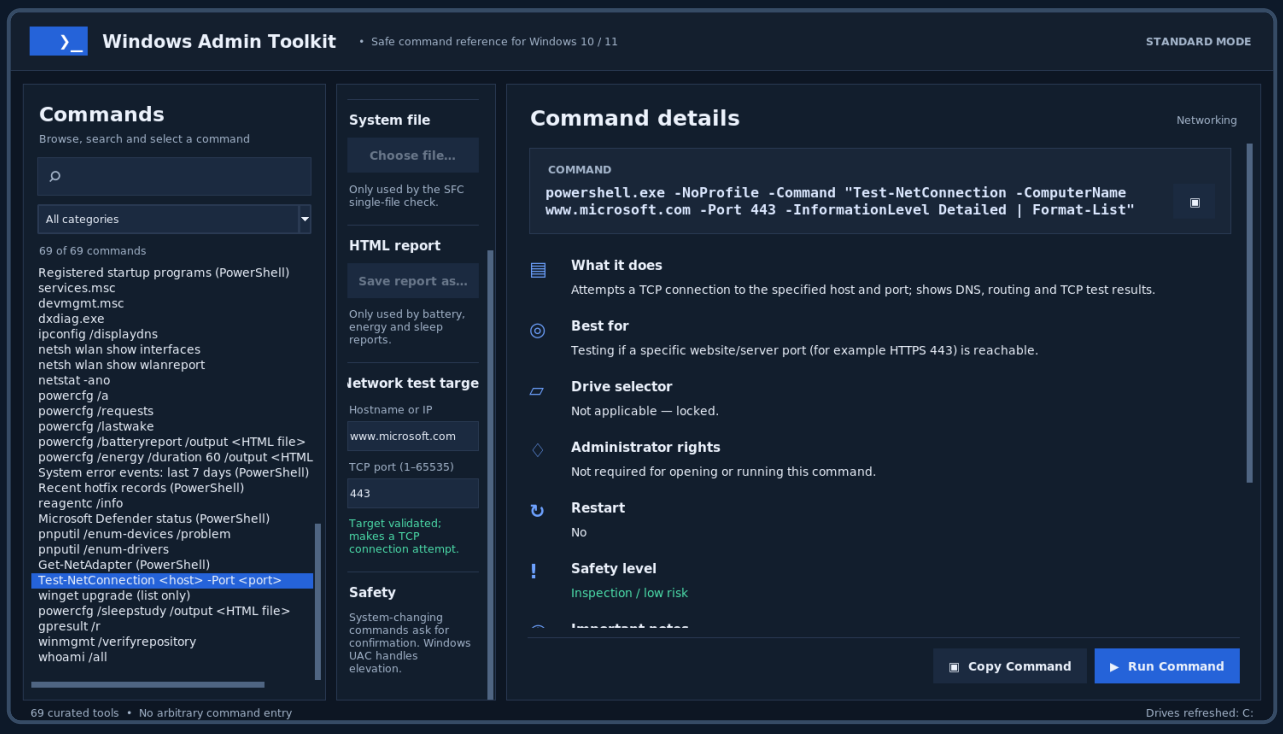
WINDOWS ADMIN TOOLKIT

The Complete User Guide & Command Handbook

Understand 69 Windows maintenance and diagnostic tools

69 built-in commands | **11 categories** | **v1.2.0 Beta**



Installation · explanations · real examples · safety · official references

October 2026 · Windows 10/11 · MIT License
github.com/DimitrisLai/Windows-Admin-Toolkit

Contents

Step-by-step usage instructions and every available Windows command.

1. Getting started and safety 3
2. Complete command reference 6
Safety key 6
Windows files (3) 6
Windows image (DISM) (4) 7
Disk & storage (11) 8
Hardware & diagnostics (5) 12
Activation & licensing (2) 13
Networking (11) 14
Logs & monitoring (9) 17
RAM & performance (5) 20
Temporary files (7) 22
Startup & drivers (6) 24
Power & battery (6) 26
Known platform restrictions 27
Accuracy and testing 28

Important: A successful launch is not proof that a command completed or repaired Windows. Always read its output.

Part 1 · Getting started and safety

Windows Admin Toolkit — User Manual

Version 1.2.0-beta.1 · 69 built-in commands · MIT · Updated 21 October 2026

1. Before you start

Windows Admin Toolkit is a **reference and launcher**, not an automatic repair program. It explains what common Windows commands do and provides a safer way to copy or open them. It does not automatically find and fix all PC problems, nor does it interpret command output.

Requires Windows 10/11, Python 3.10+ with Tkinter (standard Python for Windows). The Python source needs no extra pip packages. A built EXE is optional and must be tested on Windows.

2. Install and launch

1. Extract the ZIP archive to a local folder.
2. Install Python from python.org if necessary, including Tcl/Tk.
3. Double-click `Start_Toolkit.pyw`.
4. If `.pyw` file associations do not work, double-click `Run_Windows_Admin_Toolkit.bat` to see startup errors, or run `py -3 app.py` in the extracted directory.
5. You should see the dark GUI with a list of commands at left, drive/target controls in the middle and explanations at right.

3. Explore the window

Left — command browser: use the Search field, category dropdown, scroll wheel or Ctrl+F to find a command. Select one line; its explanation appears at right. Double-clicking a command copies it to your clipboard; double-clicking does not execute it.

Middle — drive and inputs: choose `C:` by default if detected. When a command doesn't use drive letters, the drive picker is disabled deliberately. Refresh drives after inserting/removing disks. For `sfc /scanfile` select the target file. For `powercfg` reports select an existing folder and an HTML filename. For `Test-NetConnection`, enter a host/IP and TCP port. Host/port fields are locked for other commands. The column scrolls if your screen is short.

Right — command details: the preview is generated using selected values. Underneath you see What it does, Best for, whether a drive or admin privileges are needed, restarts, safety level and important limitations. Use the Microsoft reference link when provided.

Bottom actions: Copy Command copies the exact displayed command. Run Command (or Run as Admin) requests a Windows launch. A separate terminal/tool opens; this app cannot prove whether the command succeeded.

4. Safety labels and confirmations

- Green: intended for inspection only. Even read-only tests may use network connections or generate disk/CPU load.
- Orange: writes a report, repairs files or changes some settings. A confirmation window appears.
- Red: potentially intensive or disruptive. Backup files and read the cautions before accepting.

If UAC appears, read the publisher/path and only approve an action you initiated. Cancellation should leave the system unchanged. Company policies can disallow elevation or utilities.

5. Common workflows

Windows seems corrupted: start with `DISM /CheckHealth` then `DISM /ScanHealth` or `sfc /verifyonly`. If a repair is actually needed, back up data first and use Microsoft repair instructions; `DISM /RestoreHealth` and `sfc /scannow` are changes, not harmless viewing.

Slow PC or high RAM usage: open Task Manager (`taskmgr`), Resource Monitor (`resmon.exe`), or use Top RAM Processes and physical RAM queries. Windows memory caching is normal: a RAM booster is not included.

Low disk space: run Disk Cleanup GUI or Storage Sense settings. For user TEMP, open the TEMP cleaner, preview older files, select only disposable files, review the warning and confirm. The operation permanently deletes selected files. Prefer Windows Storage Sense for broad system cleanup.

Drive issues: use `Get-PhysicalDisk`, physical-disk counters, CHKDSK without repair flags or `defrag /A`. A 'Healthy' flag does not guarantee a disk is safe; back up important information before CHKDSK repair. Never stress-test a potentially failing disk before saving data.

Network problems: check adapters, `ipconfig`, DNS cache, Wi-Fi interface, `netstat` and `Test-NetConnection` before resetting Winsock or IP. For Test-NetConnection use an approved destination (example `www.microsoft.com`, TCP 443) and read both DNS resolution and TCP test output. A failed remote port does not prove the PC has a fault.

Driver problems: open Device Manager, then `pnputil /enum-devices /problem`; inspect third-party installed driver-store packages using `/enum-drivers`. Listing packages doesn't update them.

Battery drain or sleep problems: run `powercfg /a`, `/requests`, `/lastwake`, then battery or energy report. SleepStudy works only for Modern Standby-capable devices and may not work on desktops.

Workplace policy problems: use `gpresult /r` or `whoami /all` and speak with your IT administrator rather than changing settings yourself.

6. Saving HTML reports

Battery Report, Energy Report and SleepStudy require the **Save report as…** selector. Choose an existing folder and `.html` name, then acknowledge the file-creation warning. If the report fails on the Windows computer, the terminal window normally explains why. A destination path containing spaces or non-English characters is supported as long as it does not contain disallowed shell metacharacters.

7. TEMP cleaner: precise scope

- Searches old **files** under the signed-in user's `%LOCALAPPDATA%\Temp`.
- Age options are 7, 14, 30, 60 or 90 days based on last modification time.

- Displays up to 300 oldest matching files; scan totals may be larger.
- Selects **zero** files by default; you must choose files before Delete becomes available.
- Confirms deletions; skips changed files, symlinks/junctions, files newer than the cutoff, in-use files and paths that escape the approved folder.
- **Permanently deletes** selected files; it does not use the Recycle Bin or provide an undo action.
- Doesn't remove directories or clean Downloads, browsers, Windows components, Recycle Bin or any arbitrary user-chosen TEMP directory.

Never use this first on a business PC or to recover urgently needed space. Windows Storage Sense/ Disk Cleanup are safer system-maintained alternatives.

8. Troubleshooting

Symptom	First action
App does not start	Launch `Run_Windows_Admin_Toolkit.bat` to see the Python error; check Python 3.10+ and Tkinter
No drive letters listed	Click Refresh drives; ensure a local volume is mounted; drive-specific commands remain locked until detected
Command not available	Windows edition, version, optional component or corporate policies may not provide it
Copy or Run greyed out	Choose required SFC file, report destination, valid host/port, or detected drive
UAC cancelled	Run was not approved. Do not turn off security settings just to proceed
Windows command window reports error	Read the Windows message and the Microsoft documentation linked in the UI
SleepStudy unsupported	Check `powercfg /a`; the computer may not support Modern Standby
WinGet missing	Update/install Windows Package Manager through supported Microsoft channels, or skip it

9. What has not been validated

The Linux automated tests exercise 69 commands by simulated launch request only. They do **not** establish successful outcomes on Windows, UAC behavior, or that PyInstaller builds an EXE that works on every PC. See `WINDOWS_TEST_PLAN.md` before a public stable release.

10. Privacy, support and licensing

No tracking or sign-in. Some Windows utilities can use the network. Review outputs for user names, serials, SIDs, SSIDs, hostnames and enterprise details before sharing. Bug reports should omit secrets and identifying information. The project code is MIT licensed, without warranty.

Part 2 · The 69-command handbook

Each entry gives a description, use case, warnings, example syntax, elevation/drive requirements, and official documentation. Descriptions come from the tested beta's command catalog; command execution still requires Windows validation.

Windows Admin Toolkit — Complete Command Reference

Version: 1.2.0-beta.1 · **Date:** 21 October 2026 · **License:** MIT

This document is generated from the exact 69-command `command_catalog.py` in the public beta. It is not a promise that every utility exists on every version of Windows.

Safety key

- **Inspection / low risk**: intended to inspect or list information (some may create network traffic).
- **Changes system or writes report**: may modify a setting, repair files, open an action-oriented GUI, or create/overwrite a file. The app requests confirmation.
- **Intensive / potentially disruptive**: may take a long time or reset major system settings. The app displays a stronger confirmation.

The **Run** button only confirms a Windows launch request, **not completion or success**. Read the separate terminal result.

For volume-specific commands the UI passes the currently selected drive, initially `C:` where detected; the drive selector is locked otherwise. PowerShell physical-disk checks do not use drive letters.

Windows files (3)

```
sfc /scannow
```

- **What it does:** Scans protected Windows system files and replaces corrupted versions where possible.
- **Best for:** Windows crashes, missing system files, or recurring system errors.
- **Notes / limits:** DISM /RestoreHealth may be useful first if the component store is damaged. A scan can take time.
- **Safety:** Changes settings or writes files
- **Administrator:** Yes · **Drive picker:** Locked · **Restart:** No
- **Special input:** None
- **Command example:** `sfc.exe /scannow`

- **Reference:** Microsoft documentation

```
sfc /verifyonly
```

- **What it does:** Scans protected Windows files for integrity problems without repairing them.
- **Best for:** Checking for system file problems before attempting repairs.
- **Notes / limits:** This is a read-only integrity check; any problems reported require a separate repair.
- **Safety:** Inspection / low risk
- **Administrator:** Yes · **Drive picker:** Locked · **Restart:** No
- **Special input:** None
- **Command example:** `sfc.exe /verifyonly`
- **Reference:** Microsoft documentation

```
sfc /scanfile=<file>
```

- **What it does:** Scans and, when possible, repairs one specified protected Windows system file.
- **Best for:** Checking a particular Windows system DLL or other protected file.
- **Notes / limits:** Choose an existing local file. This is intended for protected system files, not ordinary documents.
- **Safety:** Changes settings or writes files
- **Administrator:** Yes · **Drive picker:** Locked · **Restart:** No
- **Special input:** SFC file picker
- **Command example:** `sfc.exe /scanfile=C:\Windows\System32\kernel32.dll`
- **Reference:** Microsoft documentation

Windows image (DISM) (4)

```
DISM /Online /Cleanup-Image /CheckHealth
```

- **What it does:** Quickly reports whether the Windows component store was previously flagged as corrupted.
- **Best for:** A fast preliminary Windows image health check.
- **Notes / limits:** Does not perform a new deep scan and does not repair the component store.
- **Safety:** Inspection / low risk
- **Administrator:** Yes · **Drive picker:** Locked · **Restart:** No
- **Special input:** None
- **Command example:** `dism.exe /Online /Cleanup-Image /CheckHealth`
- **Reference:** Microsoft documentation

DISM /Online /Cleanup-Image /ScanHealth

- **What it does:** Performs a more thorough scan of the running Windows component store for corruption.
- **Best for:** Diagnosing Windows image health when the quick check is insufficient.
- **Notes / limits:** This may take several minutes. It scans but does not repair the image.
- **Safety:** Inspection / low risk
- **Administrator:** Yes · **Drive picker:** Locked · **Restart:** No
- **Special input:** None
- **Command example:** `dism.exe /Online /Cleanup-Image /ScanHealth`
- **Reference:** Microsoft documentation

DISM /Online /Cleanup-Image /RestoreHealth

- **What it does:** Repairs corruption in the running Windows component store using available repair sources.
- **Best for:** Fixing a damaged component store, especially before running SFC.
- **Notes / limits:** May use Windows Update to obtain repair files. Offline or restricted machines may need a /Source.
- **Safety:** Changes settings or writes files
- **Administrator:** Yes · **Drive picker:** Locked · **Restart:** No
- **Special input:** None
- **Command example:** `dism.exe /Online /Cleanup-Image /RestoreHealth`
- **Reference:** Microsoft documentation

DISM /Online /Cleanup-Image /AnalyzeComponentStore

- **What it does:** Analyzes Windows component store size and reports whether cleanup is recommended.
- **Best for:** Investigating component store disk usage without making changes.
- **Notes / limits:** A cleanup recommendation is informational; this command does not delete files.
- **Safety:** Inspection / low risk
- **Administrator:** Yes · **Drive picker:** Locked · **Restart:** No
- **Special input:** None
- **Command example:** `dism.exe /Online /Cleanup-Image /AnalyzeComponentStore`
- **Reference:** Microsoft documentation

Disk & storage (11)

chkdsk <Drive>

- **What it does:** Checks file system status on the selected local volume without requesting repairs.
- **Best for:** A preliminary review of suspected file system errors.

- **Notes / limits:** Read-only checks on active volumes may report transient discrepancies. Local volumes only.
- **Safety:** Inspection / low risk
- **Administrator:** Yes · **Drive picker:** Required · **Restart:** No
- **Special input:** None
- **Command example:** `chkdsk.exe C:`
- **Reference:** Microsoft documentation

`chkdsk <Drive> /scan`

- **What it does:** Runs an online NTFS scan of the selected local volume.
- **Best for:** Checking an NTFS drive with less disruption than an offline repair.
- **Notes / limits:** NTFS only. Some problems may still require offline repair or a restart.
- **Safety:** Inspection / low risk
- **Administrator:** Yes · **Drive picker:** Required · **Restart:** No
- **Special input:** None
- **Command example:** `chkdsk.exe C: /scan`
- **Reference:** Microsoft documentation

`chkdsk <Drive> /f`

- **What it does:** Fixes file system errors on the selected local volume.
- **Best for:** Repairing confirmed file system problems.
- **Notes / limits:** May ask to schedule a check at reboot when the volume is in use. Back up important files first.
- **Safety:** Changes settings or writes files
- **Administrator:** Yes · **Drive picker:** Required · **Restart:** May be required
- **Special input:** None
- **Command example:** `chkdsk.exe C: /f`
- **Reference:** Microsoft documentation

`chkdsk <Drive> /f /r`

- **What it does:** Checks file system errors, scans for bad sectors and attempts to recover readable data.
- **Best for:** Investigating storage read errors or suspected bad sectors.
- **Notes / limits:** Potentially very slow and intensive. Back up important data first; on a failing drive prioritize recovery.
- **Safety:** Intensive / potentially disruptive
- **Administrator:** Yes · **Drive picker:** Required · **Restart:** May be required
- **Special input:** None
- **Command example:** `chkdsk.exe C: /f /r`
- **Reference:** Microsoft documentation

`Get-PhysicalDisk (health status)`

- **What it does:** Displays disk health and operational statuses reported by Windows storage providers.
- **Best for:** A modern first look at reported physical disk health.
- **Notes / limits:** Replaces obsolete WMIC. Not a complete SMART diagnosis; reported Healthy is not proof a disk is fault-free.
- **Safety:** Inspection / low risk
- **Administrator:** Not required by the launcher · **Drive picker:** Locked · **Restart:** No
- **Special input:** None
- **Command example:** `powershell.exe -NoProfile -Command "Get-PhysicalDisk | Select-Object FriendlyName,HealthStatus,OperationalStatus,MediaType,Size | Format-Table -AutoSize"`
- **Reference:** Microsoft documentation

`Get-Volume -DriveLetter <Drive>`

- **What it does:** Shows Windows storage and file system information for the selected volume.
- **Best for:** Checking the volume file system, Windows health status, and free space.
- **Notes / limits:** Some provider fields may be unavailable for particular volumes.
- **Safety:** Inspection / low risk
- **Administrator:** Not required by the launcher · **Drive picker:** Required · **Restart:** No
- **Special input:** None
- **Command example:** `powershell.exe -NoProfile -Command "Get-Volume -DriveLetter C | Format-List DriveLetter,FileSystem,FileSystemLabel,HealthStatus,Size,SizeRemaining"`
- **Reference:** Microsoft documentation

`Disk reliability counters (PowerShell)`

- **What it does:** Displays available temperature, wear and read/write error counters from compatible physical disks.
- **Best for:** Looking for reported SSD/HDD temperature, lifetime and media-error warning signals.
- **Notes / limits:** A blank value can mean unsupported hardware or drivers, not zero; Windows storage providers vary. No complete SMART test.
- **Safety:** Inspection / low risk
- **Administrator:** Not required by the launcher · **Drive picker:** Locked · **Restart:** No
- **Special input:** None
- **Command example:** `powershell.exe -NoProfile -Command "Get-PhysicalDisk | Get-StorageReliabilityCounter | Select-Object DeviceId,Temperature,Wear,ReadErrorsTotal,WriteErrorsTotal | Format-Table -AutoSize"`
- **Reference:** Microsoft documentation

`defrag <Drive> /A /V`

- **What it does:** Analyzes the selected volume's optimization status and prints detailed information without optimizing it.
- **Best for:** Checking whether Windows considers a drive in need of optimization.
- **Notes / limits:** Administrator permission may be needed. This is not a disk integrity or SMART test.
- **Safety:** Inspection / low risk
- **Administrator:** Yes · **Drive picker:** Required · **Restart:** No
- **Special input:** None
- **Command example:** `defrag.exe C: /A /V`
- **Reference:** Microsoft documentation

`defrag <Drive> /O /V`

- **What it does:** Asks Windows to select the appropriate optimization for the volume's media type.
- **Best for:** Optimizing an HDD or requesting supported SSD retrim/maintenance using Windows defaults.
- **Notes / limits:** Can take time and generate disk activity. /O chooses appropriate behavior; do not assume it always performs traditional defragmentation.
- **Safety:** Changes settings or writes files
- **Administrator:** Yes · **Drive picker:** Required · **Restart:** No
- **Special input:** None
- **Command example:** `defrag.exe C: /O /V`
- **Reference:** Microsoft documentation

`fsutil dirty query <Drive>`

- **What it does:** Queries the volume dirty bit, which indicates Windows may need to check the file system.
- **Best for:** Investigating a volume that Windows flags for a check during startup.
- **Notes / limits:** A dirty bit does not by itself diagnose the cause. Some volumes/filesystems are unsupported; admin may be required.
- **Safety:** Inspection / low risk
- **Administrator:** Yes · **Drive picker:** Required · **Restart:** No
- **Special input:** None
- **Command example:** `fsutil.exe dirty query C:`
- **Reference:** Microsoft documentation

`manage-bde -status <Drive>`

- **What it does:** Shows BitLocker encryption, conversion and protection status for the selected volume.
- **Best for:** Verifying whether a volume is encrypted or protected before storage troubleshooting.
- **Notes / limits:** Status output can reveal identifying volume details, but this command does not display BitLocker recovery passwords.

- **Safety:** Inspection / low risk
- **Administrator:** Not required by the launcher · **Drive picker:** Required · **Restart:** No
- **Special input:** None
- **Command example:** `manage-bde.exe -status C:`
- **Reference:** Microsoft documentation

Hardware & diagnostics (5)

`mdsched.exe`

- **What it does:** Opens Windows Memory Diagnostic and offers to test RAM at the next reboot.
- **Best for:** Suspected RAM-related blue screens, instability, or random crashes.
- **Notes / limits:** Opening this tool does not immediately restart the PC; Windows asks you to choose a testing time.
- **Safety:** Changes settings or writes files
- **Administrator:** Yes · **Drive picker:** Locked · **Restart:** User chooses
- **Special input:** None
- **Command example:** `mdsched.exe`
- **Reference:** Built-in Windows tool; behavior also depends on installed components.

`systeminfo`

- **What it does:** Prints detailed information about Windows, hardware, patches, and boot time.
- **Best for:** Collecting useful system specifications for support and troubleshooting.
- **Notes / limits:** Output can contain computer names and other system details; review before sharing.
- **Safety:** Inspection / low risk
- **Administrator:** Not required by the launcher · **Drive picker:** Locked · **Restart:** No
- **Special input:** None
- **Command example:** `systeminfo.exe`
- **Reference:** Microsoft documentation

`msinfo32`

- **What it does:** Opens System Information with hardware resources, drivers, and OS details.
- **Best for:** Investigating a machine configuration through a graphical tool.
- **Notes / limits:** Read-only viewer. Information may take a moment to load.
- **Safety:** Inspection / low risk
- **Administrator:** Not required by the launcher · **Drive picker:** Locked · **Restart:** No
- **Special input:** None
- **Command example:** `msinfo32.exe`
- **Reference:** Built-in Windows tool; behavior also depends on installed components.

`winmgmt /verifyrepository`

- **What it does:** Checks the consistency of the live Windows Management Instrumentation (WMI) repository.
- **Best for:** Investigating suspected WMI repository corruption behind failing management queries.
- **Notes / limits:** Verification does not repair or reset WMI. An inconsistent result needs careful diagnosis; do not automatically reset the repository.
- **Safety:** Inspection / low risk
- **Administrator:** Yes · **Drive picker:** Locked · **Restart:** No
- **Special input:** None
- **Command example:** `winmgmt.exe /verifyrepository`
- **Reference:** Microsoft documentation

`whoami /all`

- **What it does:** Displays the security identifiers, groups and privileges in the current process access token.
- **Best for:** Understanding which account and privileges a command window actually has.
- **Notes / limits:** Results can differ between standard and elevated launches. Output includes SIDs and security-group membership; review before sharing.
- **Safety:** Inspection / low risk
- **Administrator:** Not required by the launcher · **Drive picker:** Locked · **Restart:** No
- **Special input:** None
- **Command example:** `whoami.exe /all`
- **Reference:** Microsoft documentation

Activation & licensing (2)

`slmgr /xpr`

- **What it does:** Reports whether the current Windows activation is permanent or expires.
- **Best for:** Confirming activation expiration state.
- **Notes / limits:** Uses Microsoft slmgr.vbs through cscript; does not activate Windows or change a license.
- **Safety:** Inspection / low risk
- **Administrator:** Not required by the launcher · **Drive picker:** Locked · **Restart:** No
- **Special input:** None
- **Command example:** `cscript.exe //nologo C:\Windows\System32\slmgr.vbs /xpr`
- **Reference:** Microsoft documentation

`slmgr /dlv`

- **What it does:** Shows detailed Windows licensing and activation information.
- **Best for:** Diagnosing legitimate Windows licensing issues.
- **Notes / limits:** Output includes licensing identifiers. Review before sharing publicly.
- **Safety:** Inspection / low risk
- **Administrator:** Not required by the launcher · **Drive picker:** Locked · **Restart:** No
- **Special input:** None
- **Command example:** `cscript.exe //nologo C:\Windows\System32\slmgr.vbs /dlv`
- **Reference:** Microsoft documentation

Networking (11)

`ipconfig /all`

- **What it does:** Displays network adapter addresses, DNS settings, DHCP details, and related configuration.
- **Best for:** Finding local network configuration and troubleshooting connection issues.
- **Notes / limits:** Displays local networking identifiers; review before sharing.
- **Safety:** Inspection / low risk
- **Administrator:** Not required by the launcher · **Drive picker:** Locked · **Restart:** No
- **Special input:** None
- **Command example:** `ipconfig.exe /all`
- **Reference:** Microsoft documentation

`ipconfig /flushdns`

- **What it does:** Clears the local DNS resolver cache.
- **Best for:** Troubleshooting stale or incorrect cached DNS results.
- **Notes / limits:** Does not reset network adapters or change network configuration.
- **Safety:** Changes settings or writes files
- **Administrator:** Not required by the launcher · **Drive picker:** Locked · **Restart:** No
- **Special input:** None
- **Command example:** `ipconfig.exe /flushdns`
- **Reference:** Microsoft documentation

`ipconfig /registerdns`

- **What it does:** Initiates DNS registration for eligible network adapter records.
- **Best for:** Troubleshooting DNS registration on appropriately configured networks.

- **Notes / limits:** May be irrelevant on typical home devices; DNS registration can depend on domain/network policy.
- **Safety:** Changes settings or writes files
- **Administrator:** Yes · **Drive picker:** Locked · **Restart:** No
- **Special input:** None
- **Command example:** `ipconfig.exe /registerdns`
- **Reference:** Microsoft documentation

`netsh winsock reset`

- **What it does:** Resets the Winsock catalog to its default state.
- **Best for:** Investigating persistent networking issues that may involve Winsock providers.
- **Notes / limits:** Changes networking configuration. Can affect third-party network components; reboot is usually necessary.
- **Safety:** Changes settings or writes files
- **Administrator:** Yes · **Drive picker:** Locked · **Restart:** Usually required
- **Special input:** None
- **Command example:** `netsh.exe winsock reset`
- **Reference:** Microsoft documentation

`netsh int ip reset`

- **What it does:** Resets TCP/IP settings to Windows defaults.
- **Best for:** Repairing serious network stack configuration issues.
- **Notes / limits:** May remove custom network settings (including static configuration); record settings first. Reboot afterward.
- **Safety:** Intensive / potentially disruptive
- **Administrator:** Yes · **Drive picker:** Locked · **Restart:** Usually required
- **Special input:** None
- **Command example:** `netsh.exe int ip reset`
- **Reference:** Microsoft documentation

`ipconfig /displaydns`

- **What it does:** Displays the Windows DNS Client resolver cache entries currently stored locally.
- **Best for:** Seeing whether recent DNS lookups or negative DNS cache entries exist.
- **Notes / limits:** May reveal sites or services accessed by applications; DNS entries are not a reliable browser-history record.
- **Safety:** Inspection / low risk
- **Administrator:** Not required by the launcher · **Drive picker:** Locked · **Restart:** No
- **Special input:** None
- **Command example:** `ipconfig.exe /displaydns`
- **Reference:** Microsoft documentation

`netsh wlan show interfaces`

- **What it does:** Shows Wi-Fi interface details such as connection state, radio type and current signal level.
- **Best for:** Investigating wireless signal or connection issues on a PC with Wi-Fi.
- **Notes / limits:** Requires Windows WLAN AutoConfig and compatible Wi-Fi hardware; no Wi-Fi interface means no useful data.
- **Safety:** Inspection / low risk
- **Administrator:** Not required by the launcher · **Drive picker:** Locked · **Restart:** No
- **Special input:** None
- **Command example:** `netsh.exe wlan show interfaces`
- **Reference:** Microsoft documentation

`netsh wlan show wlanreport`

- **What it does:** Generates a local HTML report of recent Windows Wi-Fi connection events and failures.
- **Best for:** Diagnosing intermittent Wi-Fi disconnects and connection errors.
- **Notes / limits:** Creates report files on disk (often under ProgramData); administrator access is recommended. Reports may contain SSIDs and device identifiers.
- **Safety:** Changes settings or writes files
- **Administrator:** Yes · **Drive picker:** Locked · **Restart:** No
- **Special input:** None
- **Command example:** `netsh.exe wlan show wlanreport`
- **Reference:** Microsoft documentation

`netstat -ano`

- **What it does:** Lists active TCP connections, listening TCP/UDP ports, numerical endpoints and owning process IDs.
- **Best for:** Investigating unexpected network listeners or finding which PID owns a connection.
- **Notes / limits:** Snapshot only; PID identifies a process but does not establish whether traffic is malicious.
- **Safety:** Inspection / low risk
- **Administrator:** Not required by the launcher · **Drive picker:** Locked · **Restart:** No
- **Special input:** None
- **Command example:** `netstat.exe -ano`
- **Reference:** Microsoft documentation

`Get-NetAdapter (PowerShell)`

- **What it does:** Lists visible network adapters with their status, reported link speed and MAC address.
- **Best for:** Checking whether Windows sees a wired or wireless adapter and whether its link is up.

- **Notes / limits:** Virtual adapters may be included; a connected link is not proof that internet access works. MAC addresses are identifying data.
- **Safety:** Inspection / low risk
- **Administrator:** Not required by the launcher · **Drive picker:** Locked · **Restart:** No
- **Special input:** None
- **Command example:** `powershell.exe -NoProfile -Command "Get-NetAdapter | Select-Object Name,InterfaceDescription,Status,LinkSpeed,MacAddress | Format-Table -AutoSize"`
- **Reference:** Microsoft documentation

```
Test-NetConnection <host> -Port <port>
```

- **What it does:** Attempts a TCP connection to the specified host and port; shows DNS, routing and TCP test results.
- **Best for:** Testing if a specific website/server port (for example HTTPS 443) is reachable.
- **Notes / limits:** Enter a trusted hostname or IP address and port (1–65535). A failed result may be caused by a firewall or remote-service policy, not necessarily by your PC. Makes outbound network traffic.
- **Safety:** Inspection / low risk
- **Administrator:** Not required by the launcher · **Drive picker:** Locked · **Restart:** No
- **Special input:** DNS/IP host and TCP port
- **Command example:** `powershell.exe -NoProfile -Command "Test-NetConnection -ComputerName www.microsoft.com -Port 443 -InformationLevel Detailed | Format-List"`
- **Reference:** Microsoft documentation

Logs & monitoring (9)

```
perfmon /rel
```

- **What it does:** Opens Reliability Monitor, showing application and Windows failures over time.
- **Best for:** Finding which crashes or failures occurred around a particular date.
- **Notes / limits:** A useful first stop for recurring crashes and application errors.
- **Safety:** Inspection / low risk
- **Administrator:** Not required by the launcher · **Drive picker:** Locked · **Restart:** No
- **Special input:** None
- **Command example:** `perfmon.exe /rel`
- **Reference:** Built-in Windows tool; behavior also depends on installed components.

```
eventvwr.msc
```

- **What it does:** Opens Windows Event Viewer to inspect system, application, and security logs.
- **Best for:** Examining warnings and failures after a specific problem.
- **Notes / limits:** Many events are routine; focus on events matching the actual time of the issue.

- **Safety:** Inspection / low risk
- **Administrator:** Not required by the launcher · **Drive picker:** Locked · **Restart:** No
- **Special input:** None
- **Command example:** `eventvwr.msc`
- **Reference:** Built-in Windows tool; behavior also depends on installed components.

`taskmgr`

- **What it does:** Opens Task Manager to view running processes and resource usage.
- **Best for:** Investigating unusually high CPU, RAM, disk, or network use.
- **Notes / limits:** Opening Task Manager is safe; avoid ending unknown processes.
- **Safety:** Inspection / low risk
- **Administrator:** Not required by the launcher · **Drive picker:** Locked · **Restart:** No
- **Special input:** None
- **Command example:** `taskmgr.exe`
- **Reference:** Built-in Windows tool; behavior also depends on installed components.

`System error events: last 7 days (PowerShell)`

- **What it does:** Shows up to 30 error-level events from the System log in the past seven days, newest first.
- **Best for:** Finding errors close to a Windows crash, hardware fault or unexpected restart.
- **Notes / limits:** No matching events is not proof the PC is healthy. Reading some event logs may require elevation.
- **Safety:** Inspection / low risk
- **Administrator:** Not required by the launcher · **Drive picker:** Locked · **Restart:** No
- **Special input:** None
- **Command example:** `powershell.exe -NoProfile -Command "Get-WinEvent -FilterHashtable @{LogName='System';Level=2;StartTime=(Get-Date).AddDays(-7)} -MaxEvents 30 | Select-Object TimeCreated,Id,ProviderName,Message | Format-List"`
- **Reference:** Microsoft documentation

`Recent hotfix records (PowerShell)`

- **What it does:** Lists up to 15 Windows hotfix records in descending installed-date order.
- **Best for:** Checking recently recorded servicing changes while troubleshooting.
- **Notes / limits:** Get-HotFix does not list every feature update, driver update or Microsoft Store update; some dates may be missing.
- **Safety:** Inspection / low risk
- **Administrator:** Not required by the launcher · **Drive picker:** Locked · **Restart:** No
- **Special input:** None
- **Command example:** `powershell.exe -NoProfile -Command "Get-HotFix | Sort-Object InstalledOn -Descending | Select-Object -First 15 HotFixID,InstalledOn,Description | Format-Table -AutoSize"`

- **Reference:** Microsoft documentation

`reagentc /info`

- **What it does:** Shows whether Windows Recovery Environment is enabled and where its recovery image is configured.
- **Best for:** Checking recovery readiness before attempting advanced Windows repairs.
- **Notes / limits:** Read-only command; some systems require administrator permissions to access all details.
- **Safety:** Inspection / low risk
- **Administrator:** Yes · **Drive picker:** Locked · **Restart:** No
- **Special input:** None
- **Command example:** `reagentc.exe /info`
- **Reference:** Microsoft documentation

Microsoft Defender status (PowerShell)

- **What it does:** Displays current antivirus and real-time protection status and signature-update date when Defender is available.
- **Best for:** Checking whether Microsoft Defender protection and signature updates are active.
- **Notes / limits:** Third-party antivirus products, security policy or Defender component availability can affect results.
- **Safety:** Inspection / low risk
- **Administrator:** Not required by the launcher · **Drive picker:** Locked · **Restart:** No
- **Special input:** None
- **Command example:** `powershell.exe -NoProfile -Command "Get-MpComputerStatus | Select-Object AntivirusEnabled,RealTimeProtectionEnabled,AntivirusSignatureLastUpdated | Format-List"`
- **Reference:** Microsoft documentation

winget upgrade (list only)

- **What it does:** Lists installed applications for which WinGet identifies available updates; does not install them.
- **Best for:** Checking application update opportunities before updating anything.
- **Notes / limits:** Requires Windows Package Manager (App Installer). Depending on source configuration, WinGet may access the network or request agreement acceptance. This is not Windows Update.
- **Safety:** Inspection / low risk
- **Administrator:** Not required by the launcher · **Drive picker:** Locked · **Restart:** No
- **Special input:** None
- **Command example:** `winget.exe upgrade`
- **Reference:** Microsoft documentation

`gpresult /r`

- **What it does:** Shows summary information about applied Group Policy settings for the current user and computer when accessible.
- **Best for:** Troubleshooting organizational restrictions and unexpected policy-controlled settings.
- **Notes / limits:** The current user context matters: elevation as a different administrator can show a different user. Some computer data may require elevation; personal PCs may have little policy information.
- **Safety:** Inspection / low risk
- **Administrator:** Not required by the launcher · **Drive picker:** Locked · **Restart:** No
- **Special input:** None
- **Command example:** `gpresult.exe /r`
- **Reference:** Microsoft documentation

RAM & performance (5)

`resmon.exe`

- **What it does:** Opens Resource Monitor to inspect per-process memory usage plus CPU, disk and networking activity.
- **Best for:** Finding applications consuming RAM and investigating resource pressure.
- **Notes / limits:** Resource Monitor is an interactive viewer; high cached RAM alone does not mean memory is faulty.
- **Safety:** Inspection / low risk
- **Administrator:** Not required by the launcher · **Drive picker:** Locked · **Restart:** No
- **Special input:** None
- **Command example:** `resmon.exe`
- **Reference:** Built-in Windows tool; behavior also depends on installed components.

`Installed RAM modules (PowerShell)`

- **What it does:** Lists memory module slots, manufacturers, module capacities in GiB, rated speeds, and part numbers as reported by firmware.
- **Best for:** Confirming installed RAM size and investigating module differences.
- **Notes / limits:** Reported module details depend on BIOS/firmware; reported Speed is not a benchmark.
- **Safety:** Inspection / low risk
- **Administrator:** Not required by the launcher · **Drive picker:** Locked · **Restart:** No
- **Special input:** None
- **Command example:** `powershell.exe -NoProfile -Command "Get-CimInstance Win32_PhysicalMemory | Select-Object BankLabel,Manufacturer,@{Name='CapacityGB';Expression={[math]::Round($_.Capacity/1GB,2)}},Speed,PartNumber | Format-Table -AutoSize"`
- **Reference:** Microsoft documentation

Installed and free RAM (PowerShell)

- **What it does:** Reports total usable and currently free physical RAM in GiB using values that Windows exposes in KiB.
- **Best for:** Checking whether Windows currently has little free physical memory.
- **Notes / limits:** Free physical memory changes constantly; Windows also intentionally uses memory for caching.
- **Safety:** Inspection / low risk
- **Administrator:** Not required by the launcher · **Drive picker:** Locked · **Restart:** No
- **Special input:** None
- **Command example:** `powershell.exe -NoProfile -Command "Get-CimInstance Win32_OperatingSystem | Select-Object @{Name='TotalRAMGiB';Expression={[math]::Round($_.TotalVisibleMemorySize/1MB, 2)}},@{Name='FreeRAMGiB';Expression={[math]::Round($_.FreePhysicalMemory/1MB,2)}} | Format-List"`
- **Reference:** Microsoft documentation

Top 20 RAM processes (PowerShell)

- **What it does:** Lists the 20 processes with the largest working sets (resident physical memory) in MiB.
- **Best for:** Identifying applications currently occupying large amounts of physical RAM.
- **Notes / limits:** Working set is not total allocated memory or proof of a leak; some protected processes may not expose details.
- **Safety:** Inspection / low risk
- **Administrator:** Not required by the launcher · **Drive picker:** Locked · **Restart:** No
- **Special input:** None
- **Command example:** `powershell.exe -NoProfile -Command "Get-Process | Sort-Object WorkingSet64 -Descending | Select-Object -First 20 ProcessName,Id,@{Name='WorkingSetMiB';Expression={[math]::Round($_.WorkingSet64/1MB,1)}} | Format-Table -AutoSize"`
- **Reference:** Microsoft documentation

perfmon.exe

- **What it does:** Opens Performance Monitor to review live counters and create performance data collection sets.
- **Best for:** Investigating intermittent resource bottlenecks with Windows performance counters.
- **Notes / limits:** Viewing is safe, but users can configure data collection inside the tool.
- **Safety:** Inspection / low risk
- **Administrator:** Not required by the launcher · **Drive picker:** Locked · **Restart:** No
- **Special input:** None
- **Command example:** `perfmon.exe`
- **Reference:** Built-in Windows tool; behavior also depends on installed components.

Temporary files (7)

`cleanmgr /d <Drive>`

- **What it does:** Opens the Disk Cleanup interface for the selected drive; you choose what categories to remove.
- **Best for:** Reclaiming disk space using Microsoft's built-in review-and-confirm cleanup UI.
- **Notes / limits:** Review the selected categories carefully; some choices can delete Recycle Bin content or other data.
- **Safety:** Changes settings or writes files
- **Administrator:** Not required by the launcher · **Drive picker:** Required · **Restart:** No
- **Special input:** None
- **Command example:** `cleanmgr.exe /d C`
- **Reference:** Microsoft documentation

`Storage Sense settings`

- **What it does:** Opens Windows Storage Sense settings to inspect automatic and manual space-cleanup options.
- **Best for:** Managing how Windows cleans temporary files and optionally the Recycle Bin or Downloads.
- **Notes / limits:** Page availability varies by Windows version. Changes made inside Settings can delete files.
- **Safety:** Inspection / low risk
- **Administrator:** Not required by the launcher · **Drive picker:** Locked · **Restart:** No
- **Special input:** None
- **Command example:** `ms-settings:storagepolicies`
- **Reference:** Microsoft documentation

`Open current-user %TEMP% folder`

- **What it does:** Opens the current user's temporary directory in File Explorer without deleting anything.
- **Best for:** Inspecting which applications have left temporary files on the PC.
- **Notes / limits:** Uses the current user TEMP path, not the drive dropdown; avoid deleting unknown or in-use files.
- **Safety:** Inspection / low risk
- **Administrator:** Not required by the launcher · **Drive picker:** Locked · **Restart:** No
- **Special input:** None
- **Command example:** `explorer.exe C:\Windows\Temp`
- **Reference:** Built-in Windows tool; behavior also depends on installed components.

Measure user TEMP folder (PowerShell)

- **What it does:** Counts accessible files inside the current user's TEMP folder and estimates their total size in GiB.
- **Best for:** Checking how much storage temporary files are occupying before cleanup.
- **Notes / limits:** Read-only, but recursion can take time; inaccessible files and junction targets are not guaranteed to be counted.
- **Safety:** Inspection / low risk
- **Administrator:** Not required by the launcher · **Drive picker:** Locked · **Restart:** No
- **Special input:** None
- **Command example:** `powershell.exe -NoProfile -Command "Get-ChildItem -LiteralPath $env:TEMP -File -Recurse -ErrorAction SilentlyContinue | Measure-Object -Property Length -Sum | Select-Object Count,@{Name='SizeGiB';Expression={[math]::Round($_.Sum/1GB,3)}} | Format-List"`
- **Reference:** Microsoft documentation

Oldest user TEMP files (PowerShell)

- **What it does:** Shows up to 20 accessible temporary files with the oldest modification dates, without deleting them.
- **Best for:** Investigating old temporary data prior to making cleanup decisions.
- **Notes / limits:** Age is based on modification time, not whether a file is safe to delete; paths and file names may contain private data.
- **Safety:** Inspection / low risk
- **Administrator:** Not required by the launcher · **Drive picker:** Locked · **Restart:** No
- **Special input:** None
- **Command example:** `powershell.exe -NoProfile -Command "Get-ChildItem -LiteralPath $env:TEMP -File -Recurse -ErrorAction SilentlyContinue | Sort-Object LastWriteTime | Select-Object -First 20 FullName,Length,LastWriteTime | Format-Table -Wrap"`
- **Reference:** Microsoft documentation

wsreset.exe

- **What it does:** Clears Microsoft Store's local cache and may reopen Microsoft Store.
- **Best for:** Investigating Store loading or cache-related problems.
- **Notes / limits:** This is specifically the Store cache; it does not clean ordinary TEMP files. Store may take time to reopen.
- **Safety:** Changes settings or writes files
- **Administrator:** Not required by the launcher · **Drive picker:** Locked · **Restart:** No
- **Special input:** None
- **Command example:** `wsreset.exe`
- **Reference:** Built-in Windows tool; behavior also depends on installed components.

`DISM /StartComponentCleanup`

- **What it does:** Removes superseded component-store files from the running Windows installation to reduce their disk usage.
- **Best for:** Reducing Windows component-store size when DISM analysis shows cleanup can help.
- **Notes / limits:** Unlike scheduled cleanup, manually running this can immediately remove superseded component versions and limit rollback of older updates. Never manually delete WinSxS.
- **Safety:** Intensive / potentially disruptive
- **Administrator:** Yes · **Drive picker:** Locked · **Restart:** No
- **Special input:** None
- **Command example:** `dism.exe /Online /Cleanup-Image /StartComponentCleanup`
- **Reference:** Microsoft documentation

Startup & drivers (6)

`Registered startup programs (PowerShell)`

- **What it does:** Lists startup entries exposed by the Win32_StartupCommand WMI class.
- **Best for:** Finding common Run-key and Startup-folder programs that launch with Windows.
- **Notes / limits:** Does not enumerate every startup mechanism, service, or scheduled task. Arguments may contain private information.
- **Safety:** Inspection / low risk
- **Administrator:** Not required by the launcher · **Drive picker:** Locked · **Restart:** No
- **Special input:** None
- **Command example:** `powershell.exe -NoProfile -Command "Get-CimInstance Win32_StartupCommand | Select-Object Name,Command,Location,User | Format-Table -Wrap"`
- **Reference:** Microsoft documentation

`services.msc`

- **What it does:** Opens the Windows Services management console for inspecting service status and startup settings.
- **Best for:** Finding a service that fails to start or consumes resources.
- **Notes / limits:** Changing or disabling critical services inside this console can break Windows features.
- **Safety:** Inspection / low risk
- **Administrator:** Not required by the launcher · **Drive picker:** Locked · **Restart:** No
- **Special input:** None
- **Command example:** `services.msc`
- **Reference:** Built-in Windows tool; behavior also depends on installed components.

`devmgmt.msc`

- **What it does:** Opens Device Manager to inspect connected devices and installed drivers.
- **Best for:** Diagnosing missing, disabled or malfunctioning hardware drivers.
- **Notes / limits:** Uninstalling devices or drivers within Device Manager may disrupt hardware until repaired.
- **Safety:** Inspection / low risk
- **Administrator:** Not required by the launcher · **Drive picker:** Locked · **Restart:** No
- **Special input:** None
- **Command example:** `devmgmt.msc`
- **Reference:** Built-in Windows tool; behavior also depends on installed components.

`dxdiag.exe`

- **What it does:** Opens DirectX Diagnostic Tool for graphics, audio, drivers and DirectX information.
- **Best for:** Investigating graphics/game/audio incompatibilities or collecting device information.
- **Notes / limits:** Reports may contain machine identifiers; review before sharing.
- **Safety:** Inspection / low risk
- **Administrator:** Not required by the launcher · **Drive picker:** Locked · **Restart:** No
- **Special input:** None
- **Command example:** `dxdiag.exe`
- **Reference:** Built-in Windows tool; behavior also depends on installed components.

`pnputil /enum-devices /problem`

- **What it does:** Lists Plug and Play devices that currently report a device problem code.
- **Best for:** Investigating unrecognized hardware, disabled devices or drivers that did not start.
- **Notes / limits:** Available on Windows 10 version 1903 and newer. A problem code identifies a device state, not necessarily the root cause. Does not repair devices.
- **Safety:** Inspection / low risk
- **Administrator:** Yes · **Drive picker:** Locked · **Restart:** No
- **Special input:** None
- **Command example:** `pnputil.exe /enum-devices /problem`
- **Reference:** Microsoft documentation

`pnputil /enum-drivers`

- **What it does:** Lists third-party driver packages stored in the Windows driver store, including published INF names.
- **Best for:** Checking OEM driver packages and troubleshooting driver-version conflicts.
- **Notes / limits:** Shows driver-store packages, not a complete list of currently active device drivers. Does not install, remove or update drivers.

- **Safety:** Inspection / low risk
- **Administrator:** Yes · **Drive picker:** Locked · **Restart:** No
- **Special input:** None
- **Command example:** `pnputil.exe /enum-drivers`
- **Reference:** Microsoft documentation

Power & battery (6)

```
powercfg /a
```

- **What it does:** Reports sleep states available and unavailable on the current computer, with reasons.
- **Best for:** Checking whether Sleep, Hibernate or Modern Standby is supported.
- **Notes / limits:** Capabilities depend on device firmware, drivers and current Windows configuration.
- **Safety:** Inspection / low risk
- **Administrator:** Not required by the launcher · **Drive picker:** Locked · **Restart:** No
- **Special input:** None
- **Command example:** `powercfg.exe /a`
- **Reference:** Microsoft documentation

```
powercfg /requests
```

- **What it does:** Lists currently reported processes, drivers and services requesting that Windows stay awake.
- **Best for:** Troubleshooting sleep or display-off problems.
- **Notes / limits:** The output is a live snapshot; an empty list does not rule out all sleep problems. Admin may reveal more.
- **Safety:** Inspection / low risk
- **Administrator:** Not required by the launcher · **Drive picker:** Locked · **Restart:** No
- **Special input:** None
- **Command example:** `powercfg.exe /requests`
- **Reference:** Microsoft documentation

```
powercfg /lastwake
```

- **What it does:** Displays the last wake source that Windows was able to record.
- **Best for:** Finding why a PC recently woke up unexpectedly.
- **Notes / limits:** May show no useful source, depending on firmware, sleep state and Windows logging.
- **Safety:** Inspection / low risk
- **Administrator:** Not required by the launcher · **Drive picker:** Locked · **Restart:** No
- **Special input:** None
- **Command example:** `powercfg.exe /lastwake`
- **Reference:** Microsoft documentation

```
powercfg /batteryreport /output <HTML file>
```

- **What it does:** Generates an HTML battery usage and capacity report at a file location you choose.
- **Best for:** Reviewing battery capacity trends on laptops or tablets.
- **Notes / limits:** Most desktops without batteries have no meaningful battery history. This creates or overwrites the selected HTML file and may reveal device and usage information.
- **Safety:** Changes settings or writes files
- **Administrator:** Not required by the launcher · **Drive picker:** Locked · **Restart:** No
- **Special input:** HTML report destination
- **Command example:** `powercfg.exe /batteryreport /output C:\Users\Public\report.html`
- **Reference:** Microsoft documentation

```
powercfg /energy /duration 60 /output <HTML file>
```

- **What it does:** Runs a 60-second Windows power-efficiency analysis and writes an HTML report to a selected path.
- **Best for:** Finding power-efficiency problems when idle and troubleshooting battery drain.
- **Notes / limits:** Run while the system is otherwise idle. Requires admin, waits about 60 seconds and may flag benign warnings.
- **Safety:** Changes settings or writes files
- **Administrator:** Yes · **Drive picker:** Locked · **Restart:** No
- **Special input:** HTML report destination
- **Command example:** `powercfg.exe /energy /duration 60 /output C:\Users\Public\report.html`
- **Reference:** Microsoft documentation

```
powercfg /sleepstudy /output <HTML file>
```

- **What it does:** Generates a report of Modern Standby sleep sessions, power usage and activity where supported.
- **Best for:** Investigating battery drain or poor Modern Standby behavior during sleep.
- **Notes / limits:** Requires Modern Standby-capable hardware and administrator permissions. Creates or overwrites the chosen HTML report; unsupported systems may report an error.
- **Safety:** Changes settings or writes files
- **Administrator:** Yes · **Drive picker:** Locked · **Restart:** No
- **Special input:** HTML report destination
- **Command example:** `powercfg.exe /sleepstudy /output C:\Users\Public\report.html`
- **Reference:** Microsoft documentation

Known platform restrictions

- PnPUtil `/enum-devices /problem` needs Windows 10 1903 or newer.
- WinGet may not be installed and may contact remote package sources or request source agreement. Listing does not install updates.

- SleepStudy requires Modern Standby hardware; PowerShell modules and device-specific counters may be missing.
- Disk and networking repair tools may need elevation, restart, backups and a maintenance window.
- `Test-NetConnection` sends network traffic to the user-selected host/port; it does not change system configuration.
- Windows 10 reaches end of general support in October 2025, but the toolkit can still launch its existing built-in utilities on supported configurations.

Accuracy and testing

Descriptions and syntaxes were checked against Microsoft command documentation, and commands were tested through the toolkit's command builder and mocked Windows launch requests. Actual execution on Windows machines is still outstanding. See `AUDIT_REPORT.md` and `WINDOWS_TEST_PLAN.md`.